

Exam Date & Time: 15-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Network Security [NPT_4148]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) A student wants to send the message "MEET AT GATE" to a friend using a Caesar Cipher with a shift key of 4.
 - A) a) Outline the encryption and decryption algorithms, including the role of the key in both processes. (5)
b) Encrypt the message step by step using the Caesar Cipher technique, showing all intermediate steps clearly.
 - B) A company designs a simple authentication protocol (AP2.0) where the client sends a nonce to the server during login to avoid replay attacks.
 - a) Apply how nonce-based authentication is used to prevent replay attacks during login.
 - b) Examine the sequence of message exchanges between the client and server in the AP2.0 protocol. (5)
 - c) Use your knowledge of cryptographic randomness to identify one security issue if the nonce is not truly random, and propose an appropriate measure
- 2) A multinational company uses S/MIME to secure internal email exchanges integrated with MIME messages.
 - A) a) Analyze how S/MIME extends MIME to provide message integrity and authentication. (5)
b) Compare PGP and S/MIME in terms of key management and trust model.
c) Analyze why large enterprises often prefer S/MIME over PGP for business communication.
 - B) An organization wants to securely connect remote employees to its internal network through a VPN tunnel.
 - a) Examine how Tunnel Mode works in IPSec and how it differs from Transport Mode. (5)
 - b) Demonstrate how private networks and VPNs together ensure secure remote access.
 - c) Show one real-world advantage of using Tunnel Mode in enterprise VPNs.
- 3) An organization uses WPA (with TKIP) for Wi-Fi security but wants to understand how session keys are securely exchanged.
 - A) a) Outline the four-way handshake process used in 802.11i and how it ensures mutual authentication. (5)
b) Breakdown the generation of master key in TKIP.
c) Analyze how replay attack prevention is achieved in TKIP.

- B) A telecom provider upgrades from 2G to 3G (UMTS) to improve security and performance.
 a) Demonstrate the 3G architecture, showing the role of the EPS/UTRAN components.
 b) Assess the Authentication and Key Agreement (AKA) process in 3G and how it improves over GSM.
 c) Report the purpose of message integrity and key hierarchy in 3G security. (5)
- 4) A business suffers a SYN-flood attack that exhausts the capacity of its web server's TCP backlog.
 A) a) Analyze the normal TCP three-way handshake and the normal TCP connection termination sequence.
 b) Analyze how a SYN-flood disrupts this process and outline how SYN cookies (TCP cookie defense) mitigate the attack; include what the server avoids storing when cookies are used.
 c) Infer why SYN cookies may be suboptimal for some legitimate connections. (5)
- B) An ISP observes that its customers are suffering from data exfiltration and spoofed IP attacks. To address this, the ISP considers implementing egress filtering and distribution route filtering.
 a) Demonstrate how egress filtering and distribution route filtering work to prevent network attacks.
 b) Apply your understanding of routing security to describe how route filtering works, and demonstrate how shortcomings in distributed route filtering could impact legitimate network traffic. (5)
- 5) A merchant accepts Bitcoin payments online and wants to ensure all received transactions are valid and irreversible.
 A) a) Analyze the process of a Bitcoin transaction, describing inputs, outputs, and unspent transaction outputs (UTXO).
 b) Infer how confirmation and number of confirmations help protect against double spending attacks.
 c) Outline what might happen if a user attempts to spend the same Bitcoin in two different transactions before confirmation? (5)
- B) A healthcare company is using cloud computing to store patient data and wants to ensure data privacy even while processing sensitive information.
 a) Analyze the Cloud Computing Reference Architecture, highlighting the roles of cloud consumer, provider, and broker.
 b) Breakdown the concept of Fully Homomorphic Encryption (FHE) and how it protects data in cloud environments.
 c) Outline a simple FHE scheme and mention one potential application in cloud data analytics. (5)

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Event Name	Subject
058-END SEMSTER-ODD-2025	NPT_4148:Network Security

#	Course Outcome	Description	Marks
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1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	15
4	4	CO 4	15
Total			50

* NA - CO Not Selected

Event Name	Subject
058-END SEMSTER-ODD-2025	NPT_4148:Network Security

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	4	20(40%)
4	Analysis	6	30(60%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)